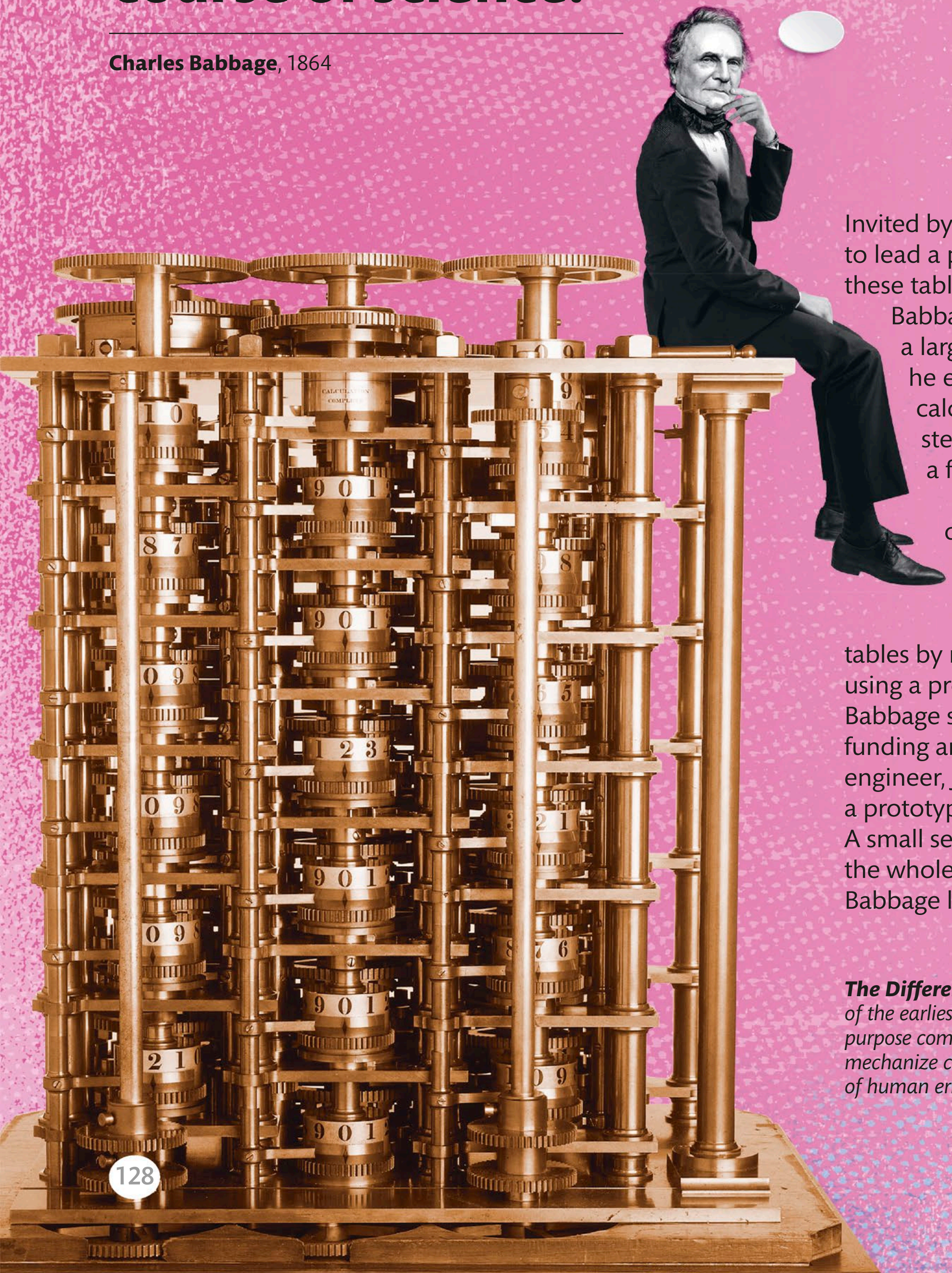
“As soon as an **Analytical Engine exists**, it will necessarily **guide the future course of science.”**

Charles Babbage, 1864



Invited by the Astronomical Society to lead a project to improve a set of these tables for *The Nautical Almanac*, Babbage found that they contained a large number of mistakes. In 1821, he exclaimed, “I wish to God these calculations had been executed by steam!” He determined to create a foolproof calculating machine. Babbage presented his first design for a Difference Engine in 1822. Powered by cranking a handle, it would formulate and print the mathematical tables by making complex calculations using a process of repeated addition. Babbage secured initial government funding and began work with his engineer, Joseph Clement, to build a prototype: Difference Engine No.1. A small section, about one-seventh of the whole, was finished in 1832, and Babbage liked to demonstrate it to

The Difference Engine No.1 *is one of the earliest conceptions of a general-purpose computer. Babbage aimed to mechanize calculation, reducing the risk of human error.*